

$$Z[J] = \langle 0 | e^{-S_F} e^{\int d^4x J\phi} | 0 \rangle = \int D\phi e^{-S_0 + J\phi - S_F}$$

$$\text{where } Z_0[J] = e^{\frac{1}{2}J \cdot \Delta J} \langle 0 | 0 \rangle = e^{\frac{1}{2}J \Delta J}$$

Examples: $G_n^{(0)}(x_1, \dots, x_n) = \frac{\delta}{\delta J_1} \dots \frac{\delta}{\delta J_n}$

(i) Consider the free theory:

$$G_2^{(0)}(x_1) \stackrel{\leftarrow \text{free theory}}{=} \frac{\delta}{\delta J_1} e^{\frac{1}{2}J \Delta J} \Big|_{J=0} = \Delta \cdot J_{x_1} e^{\frac{1}{2}J \Delta J} = 0$$

$$\text{and so we find: } n = 2k+1, \quad k \in \mathbb{Z}$$

$$G_{2k+1}^{(0)}(x_1, \dots, x_{2k+1}) = 0$$

which is what we find for $Z_0[N(\phi_1 \dots \phi_n)]|_0 = 0$
 $n = 2k+1$.

* For $n=2$:

$$G_2^{(0)}(x_1, x_2) = \frac{\delta}{\delta J_1} \frac{\delta}{\delta J_2} e^{\frac{1}{2}J \Delta J} = (\Delta_{12} + \Delta_{1\omega} J^\omega \Delta_{2\omega} J^\omega) |_{J=0}$$

at $J=0$:

$$G_2^{(0)}(x_1, x_2) = \Delta(x_1, x_2)$$

* Interacting theory: $\lambda/\Gamma = \lambda \phi^3/3!$

Note: Previously I calculated

$Z[J]$ up to level λ^3 but to calculate n -pt f^{ns} we must have

$$G_n^{(ns)}(x_1, \dots, x_n) = \frac{\delta}{\delta J_1} \dots \frac{\delta}{\delta J_n} Z^{(ns)}[J] \Big|_{J=0}$$

$$Z[J] = (1 + Z^{(1)}[J] + \frac{1}{2!} Z^{(2)}[J] + \dots)$$

$$Z^{(1)}[J] \sim \lambda, \quad Z^{(2)}[J] \sim \lambda^2 \dots$$

$$Z[J] = (1 - \int d^4x V(\frac{\delta}{\delta J}) Z_0[J] + (\quad) \dots)$$

$$S_0 = \int d^4x (\frac{\delta}{\delta J}) Z_0[J] = Z^{(1)}[J]$$

$$Z^{(1)}[J] = -\frac{\lambda}{3!} (3 \Delta_{xx} \Delta_{xw} J^w + (\Delta_{xw} J^w)^3) e^{\frac{1}{2} J \Delta J}$$

To calculate n-pt^h fⁿ at first level:

$$G_{2n}^{(1)}(x_1, \dots, x_n) = 0$$

$$G_1^{(1)}(x_1) = \langle \phi_{x_1} \rangle_J = \frac{\delta}{\delta J_{x_1}} Z^{(1)}[J] \Big|_{J=0} = -\frac{\lambda}{2} \int d^d x \Delta_{xx} \Delta_{xx}$$

$$G_3^{(1)}(x_1, x_2, x_3) = \langle \phi_{x_1} \phi_{x_2} \phi_{x_3} \rangle_J = \frac{\delta}{\delta J_{x_1}} \dots \frac{\delta}{\delta J_{x_3}} Z^{(1)}[J] \Big|_{J=0}$$

$$= -\lambda \int d^d x \left\{ \Delta_{x_1} \Delta_{x_2} \Delta_{x_3} + \frac{1}{2} \Delta_{xx} (\Delta_{x_1} \Delta_{x_2} + \Delta_{x_2} \Delta_{x_3} + \Delta_{x_3} \Delta_{x_1}) \right\}$$



at level λ^2 :

$$Z^{(2)}[J] \Big|_{J=0} = \frac{\lambda^2}{2^3} \Delta_{xx} \Delta_{xy} \Delta_{yy} + \frac{\lambda^2}{2 \times 3!} \Delta_{xy}^3$$



Wicks thm: Second form:

Lemma 1 (Coleman): Consider two fⁿs:

$$f(x) = f(x_1, \dots, x_n), \quad g(y) = g(y_1, \dots, y_n)$$

Then:

$$f\left(\frac{\partial}{\partial x}\right) g(x) = g\left(\frac{\partial}{\partial y}\right) \left(f(y) e^{x \cdot y} \right) \Big|_{y=0}$$

Proof: $f(x) = \int \frac{d^d p}{(2\pi)^3} e^{-ipx} \hat{f}(p)$

and so we can simply consider

$$f(x) = e^{i x \cdot p} \quad \text{and} \quad g(y) = e^{i y \cdot p}$$

for the proof.

$$\begin{aligned} \text{LHS: } f\left(\frac{\partial}{\partial x}\right) g(x) &= e^{a \frac{\partial}{\partial x}} e^{bx} = \sum_n \frac{1}{n!} \left(a \frac{\partial}{\partial x}\right)^n e^{bx} \\ &= \sum_n \frac{1}{n!} (a^n b^n) e^{bx} = e^{a^n b^n} e^{bx} \\ &= e^{b(a+x)} \end{aligned}$$

$$\text{RHS: } e^{b \frac{\partial}{\partial y}} (e^a e^{xy}) \Big|_{y=0} = e^{b(x+a)}$$

* Applied to Wick's thm $x \rightarrow \tau$

$$e^{-\int d^4x V(\frac{\delta}{\delta \tau})} e^{-\frac{1}{2} \tau \Delta \tau} \quad y \rightarrow \phi$$

$$f\left(\frac{\delta}{\delta \tau}\right) [g(\tau)] = g\left(\frac{\partial}{\partial \phi}\right) \left\{ f(\phi) e^{\tau \phi} \right\}_{\phi=0}$$

$$Z[\tau] = e^{\frac{1}{2} \frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi}} \left\{ e^{-\int d^4x V(\phi) + \tau \phi} \right\}_{\phi=0}$$

Feynman Rules in x -space:

$$V(\phi) = \lambda \phi^p$$

ex: $p=4$, take $n=2$ $x_1, x_2 \dots$
at order λ :

$$x_1 \bullet \quad \bullet x_2$$



$$\text{V} + \text{O} + \text{perms.}$$